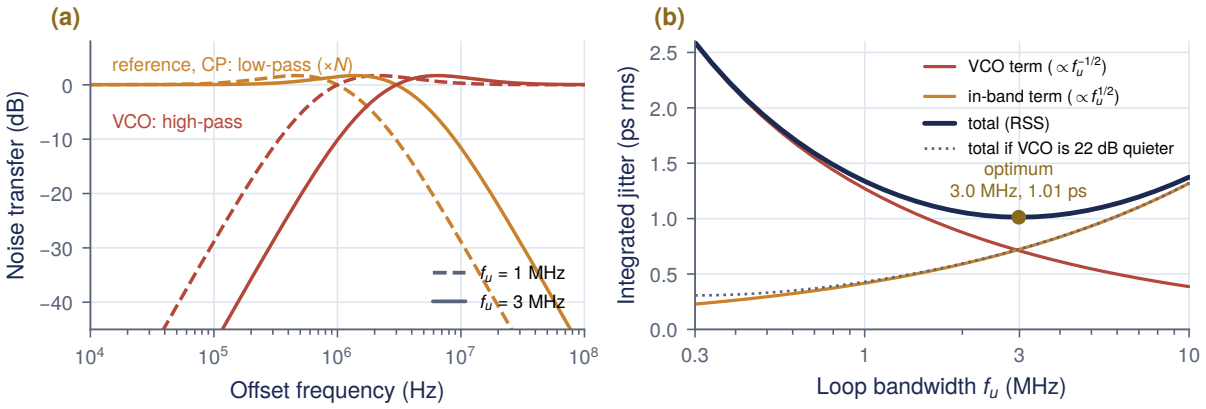
(a)
reference, CP: low-pass (×N)
VCO: high-pass
Noise transfer (dB)
0
−10
−20
−30
−40
f_u = 1 MHz
f_u = 3 MHz
10^4
10^5
10^6
10^7
10^8
Offset frequency (Hz)
(b)
VCO term ($\propto f_u^{-1/2}$)
in-band term ($\propto f_u^{1/2}$)
total (RSS)
total if VCO is 22 dB quieter
optimum
3.0 MHz, 1.01 ps
Integrated jitter (ps rms)
2.5
2.0
1.5
1.0
0.5
0.0
0.3
1
3
10
Loop bandwidth f_u (MHz)